Fluid Mechanics & Hydraulic Machinery Notes

Department: Mechanical Engineering | Semester: 4 | Code: ME402
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1. Fluid Statics & Pascal's Principle

Pressure in a static fluid acts equally in all directions: P = rho * g * h. Buoyancy force equals the weight of the fluid displaced according to Archimedes.

2. Bernoulli's Equation for Incompressible Flow

Conservation of mechanical energy along a streamline: P/rho + v^2/2 + g*z = constant. Applies to Venturi meters and Pitot tubes.

3. Boundary Layer Theory & Drag Force

Prandtl's boundary layer concept explains friction drag and flow separation over aerofoils and immersed cylinders.